



Ampere Shower Power Speaker

Listening to music while in the shower is nothing new, lots of people do it. However, unless you have a decent speaker, that's also water resistant, it's not exactly a good time. If it's directly through our phones, you can be sure that most of us have them a mile away from water, just to be on the safe side of things. So, what's the solution? Ampere has come up with the Shower Power. This speaker easily attaches to most showerheads, and is actually powered by flowing water. Also, it's made with 100 percent recycled plastic. On the inside of the Shower Power you have a small propeller which spins when water flows through it. This powers a mini generator that charges the battery. On a full charge, the Shower Power will last up to 16 hours, and it'll work even when the shower's off. Ampere also claims they haven't compromised on sound quality. The Shower Power uses a proprietary Sound Wave amplifier that plays music 360 degrees around the speaker, supposedly delivering evenly distributed, full-room sound. You can check the Shower Power out on Kickstarter, where you can also pre-order the speakers. It's expected to start shipping in January.

Raspberry Pi 400

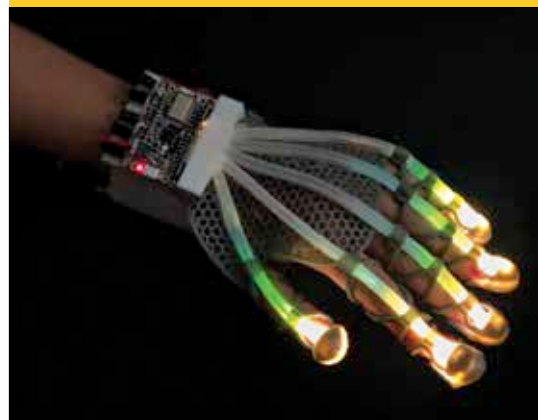
The latest variant of the Raspberry Pi, the Raspberry Pi 400 looks like a keyboard. But it's a keyboard with a complete personal computer built into it. Inside, you've got a quad-core 64-bit processor, 4GB of RAM, wireless networking, dual-display output, and 4K video playback, as well as a 40-pin GPIO header. All of that built into a rather compact keyboard. Pretty impressive. The Raspberry Pi 400 uses a purpose-built board based on the Raspberry Pi 4. It uses the same processor as the Pi 4. The Pi 400 has specially designed thermals



to keep the computer cool and silent while you work. The personal computer kit for the Pi 400 comes with a mouse, power supply, micro HDMI to HDMI cable and an SD card preloaded with Raspberry Pi OS. If you don't want the PC kit, you can of course get the Pi 400 as a standalone unit as well, and use your own accessories with it.

Cornell Skin Sensor

Virtual Reality has been around for a while now, and there are many offerings that increase immersion, by allowing users to feel and sense more of the virtual world. Cornell scientists are taking this immersion to another level with the development of a stretchable skin sensor that uses fibre optics, which allows VR users to have a sense of touch. For now, there's a prototype glove, where each finger has a stretchable lightguide (as can be seen in the image). These



lightguides combine the transparent polyurethane core and an LED-linked core which has absorbent dyes loaded inside. When you bend your fingers and deform the lightguides, or when you feel pressure in VR, the dyes light up and register where the sensation will be felt. The current prototype setup is a 3D-printed one-off that works with Bluetooth, and basic circuitry, but the potential uses are limitless. For instance, a future where you can touch and grab virtual objects and actually feel the sensation in your hands is not far off. Additionally, another use that might come out of these gloves is the ability to give robots a sense of touch. This would allow them to better react to their direct environment, and handle delicate objects, delicately.